

Physical Chemistry (II)

Lecture 04

Principles of Quantum Mechanics

Particle in a Box



In the Last Class

1. Wavefunction and its interpretation

2. Quantum measurement

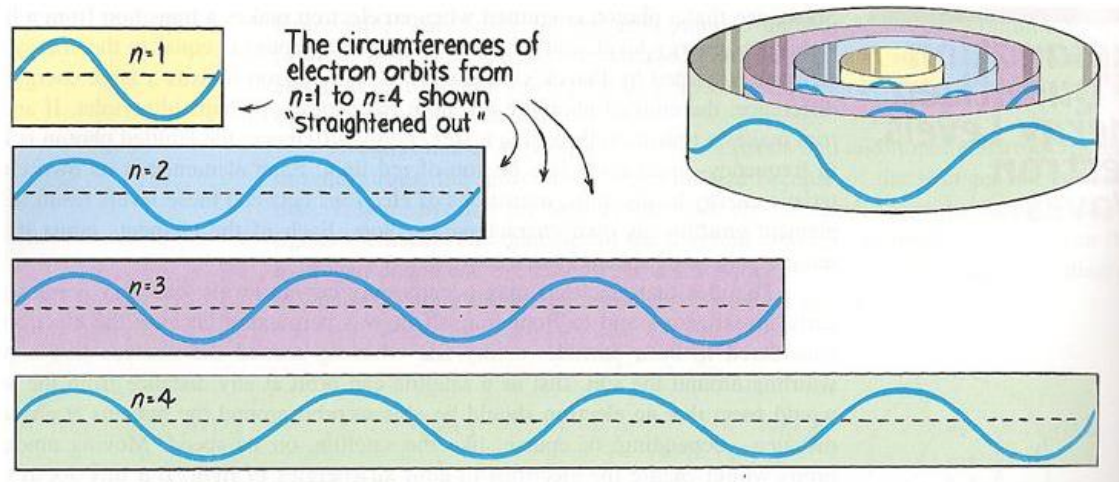
- Observables and operators
- Measurement and collapse of wave function ←
- Uncertainty principle

3. Time evolution of wavefunction: Schrödinger equation

Last class

Wavefunction

The state of a quantum-mechanical particle is represented by a **certain complex function** that behaves like a **wave**



(Image: <https://chemistry.stackexchange.com/questions/116840/why-do-solutions-of-electron-in-a-box-and-in-a-ring-predict-coefficients-for-l>)

Last class

“Amplitude” of a Wavefunction

For a wavefunction $\Psi(x, t)$,
its square $|\Psi(x, t)|^2$ gives the **probability**
of finding the particle at point x at time t
(Born's statistical interpretation)

What really matters is the **PROBABILITY!**



Observables and Operators

- Every **CM observable** has a corresponding **QM operator**

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$

- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

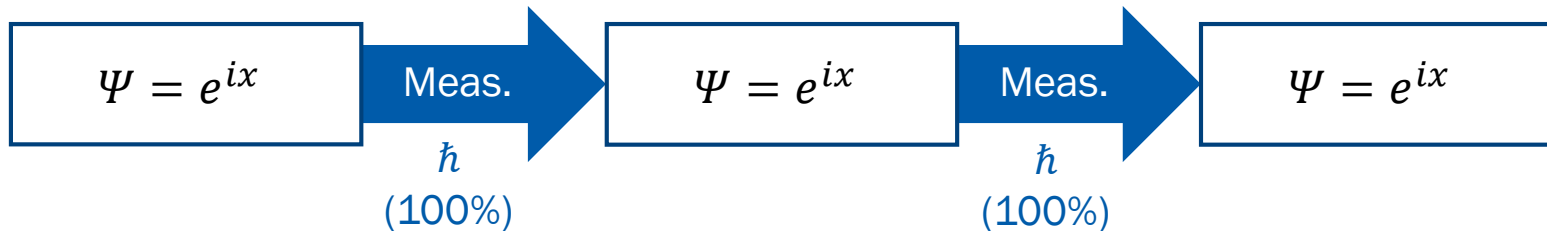
$\hbar/2\pi$
↓
○

- System energy operator (Hamiltonian):

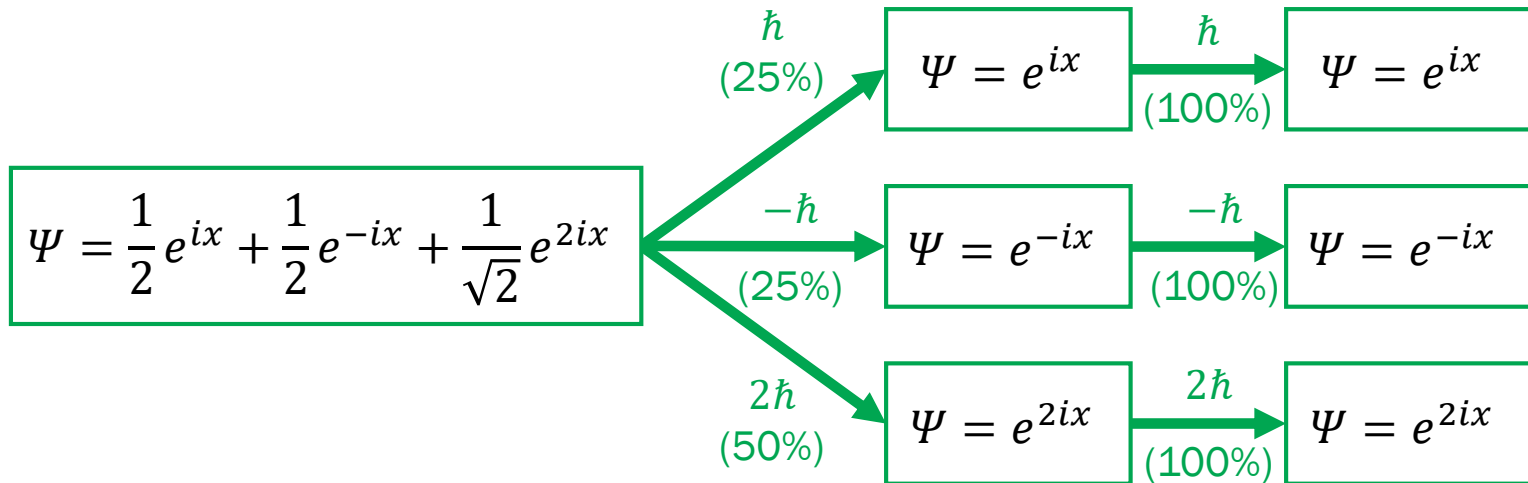
$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z})$$

Quantum Measurement

- If the wavefunction is the eigenfunction of the operator



- Otherwise



Review: Statistics

- Expected value

$$\langle g \rangle = \sum_{i=1}^{\infty} p_i g_i$$

- Variance

$$\sigma_g^2 = \langle (g - \langle g \rangle)^2 \rangle = \langle g^2 \rangle - \langle g \rangle^2$$

- Standard deviation

$$\sigma_g = (\sigma_g^2)^{1/2} = \sqrt{\langle g^2 \rangle - \langle g \rangle^2}$$

Expectation Value

For the eigenfunctions of the operator $\hat{A}$,

$$\hat{A}f_k = a_k f_k,$$

if the wavefunction of the system is

$$\Psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the expectation value of the observable A is

$$\begin{aligned}\langle A \rangle &= a_1 P_1 + a_2 P_2 + \cdots + a_n P_n \\ &= a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2\end{aligned}$$

RS 2.6

M 4.3

Expectation Value

$$\langle A \rangle = a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2$$

Using the orthogonality (and normalization), we can show

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi \, d\tau$$



Example 4-4

Calculate the average energy of the system in a state represented by

$$\Psi(x) = \left(\frac{30}{a^5}\right)^{1/2} x(a-x), \quad 0 \leq x \leq a$$

The average energy is the expectation value of energy:

$$\langle E \rangle = \int_0^a \Psi^*(x) \hat{H} \Psi(x) dx = \int_0^a \Psi^*(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \Psi(x) dx$$

$$\langle E \rangle = -\frac{15\hbar^2}{ma^5} \int_0^a x(a-x) \frac{d^2}{dx^2} [x(a-x)] dx$$

$$\langle E \rangle = \frac{30\hbar^2}{ma^5} \int_0^a x(a-x) dx = \frac{5\hbar^2}{ma^2}$$

Variance of QM Measurement

$$\sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$$

where

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, d\tau$$

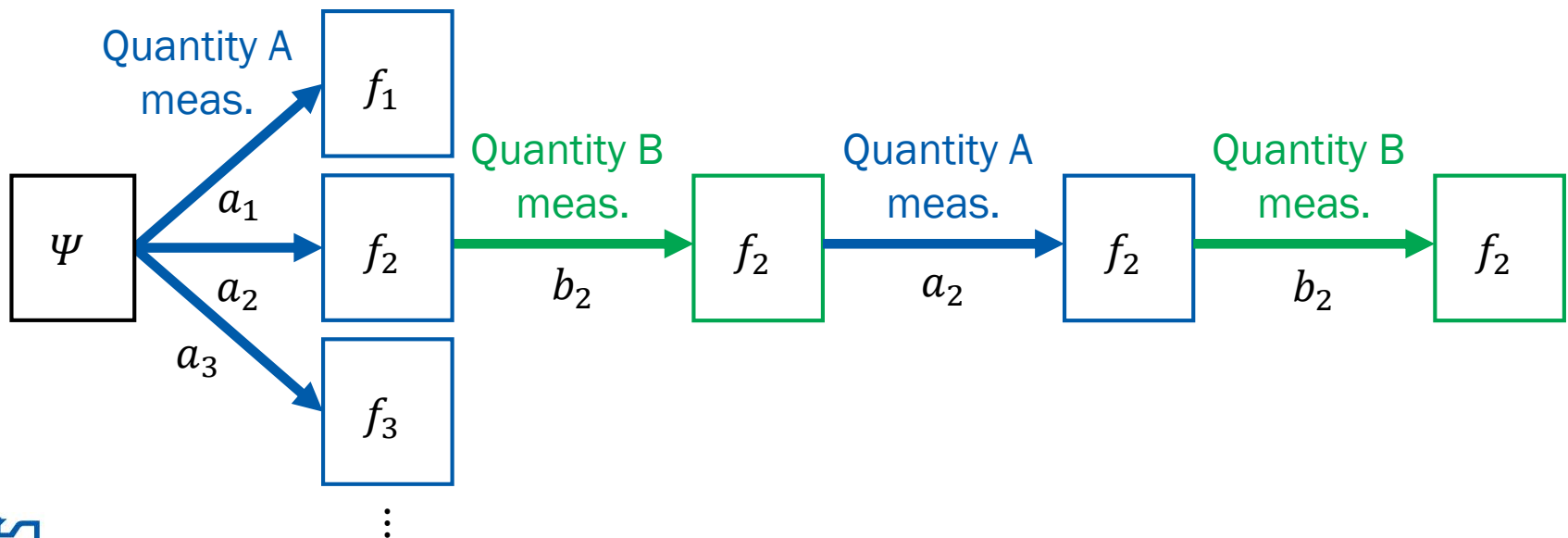
$$\langle A^2 \rangle = \int \psi^* \hat{A} \hat{A} \psi \, d\tau$$

- Standard deviation: $\sigma_A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$

M 4.7 Measurement of Two Observables

If two QM operators share their eigenfunctions ...

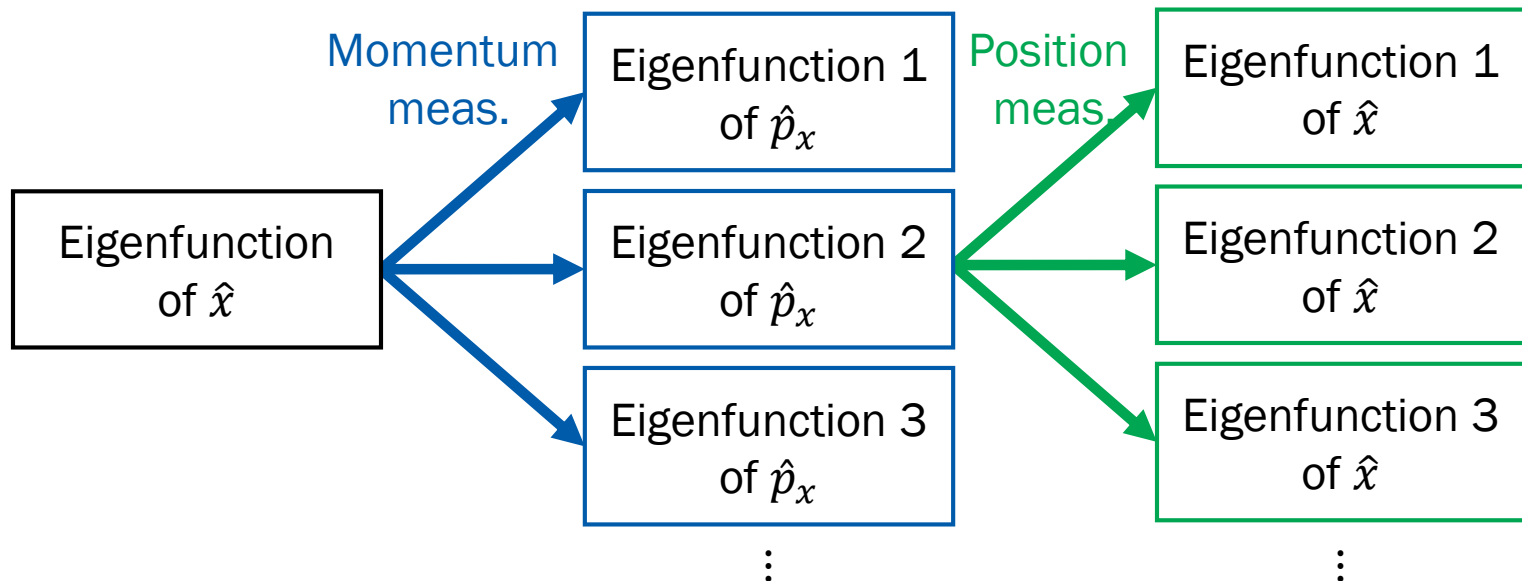
$$\hat{A}f_k = a_k f_k \text{ and } \hat{B}f_k = b_k f_k$$



M 4.4 Measurement of Two Observables

The eigenfunction of one QM operator may not be the eigenfunction of another QM operator

(e.g. eigenfunction of $\hat{x} \neq$ eigenfunction of $\hat{p}_x$)



Uncertainty Principle

For two QM operators that do not share the eigenfunctions,

It is impossible to simultaneously determine
the corresponding observables

- Position-momentum uncertainty (Gen Chem version)

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

Uncertainty Principle

For two QM operators that do not share the eigenfunctions,

It is impossible to simultaneously determine
the corresponding observables

- Position-momentum uncertainty (formal version)

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2}$$

Operator Multiplication

- Associative property

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property

$$\hat{A}\hat{B} = \hat{B}\hat{A} ? \rightarrow \text{Not necessarily}$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to **commute**
- **Commutator**

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Generalized Uncertainty Principle

$$\sigma_A^2 \sigma_B^2 \geq -\frac{1}{4} \langle [\hat{A}, \hat{B}] \rangle^2$$

e.g. position-momentum uncertainty

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = x \frac{\hbar}{i} \frac{\partial}{\partial x} - \frac{\hbar}{i} \frac{\partial}{\partial x} x = i\hbar$$

$$\sigma_x^2 \sigma_{p_x}^2 \geq -\frac{1}{4} \langle i\hbar \rangle^2 = \left(\frac{\hbar}{2} \right)^2$$

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2}$$

Time Evolution of Wavefunction

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, z, t) = \hat{H} \Psi(x, y, z, t)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}, t)$$

(Time-dependent Schrödinger equation)

Time-Independent Potential

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, z, t) = \hat{H} \Psi(x, y, z, t)$$

If the potential is independent of time, we assume

$$\Psi(x, y, z, t) = \psi(x, y, z)T(t)$$

and divide both sides by $\Psi(x, y, z, t)$ to obtain

$$\frac{i\hbar}{T} \frac{\partial T}{\partial t} = \frac{\hat{H}\psi}{\psi}$$

(separation of variables)

Time-Independent Equation

$$\frac{i\hbar}{T} \frac{\partial T}{\partial t} = \frac{\hat{H}\psi}{\psi}$$

Letting both sides equal to a constant E , we obtain

$$\hat{H}\psi(x, y, z) = E\psi(x, y, z)$$

(time-independent Schrödinger equation)

$$\text{and } T(t) = e^{-iEt/\hbar}$$

Solutions

- Time-independent Schrödinger equation

$$\hat{H}\psi(x, y, z) = E\psi(x, y, z)$$

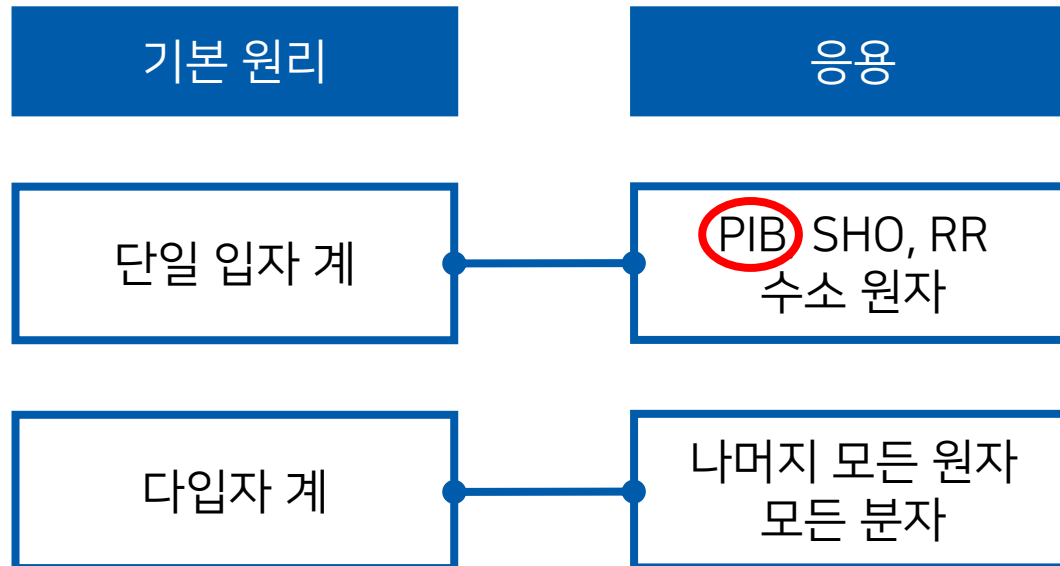
→ Eigenvalue equation for the Hamiltonian

- Temporal solution: $T(t) = e^{-iEt/\hbar}$

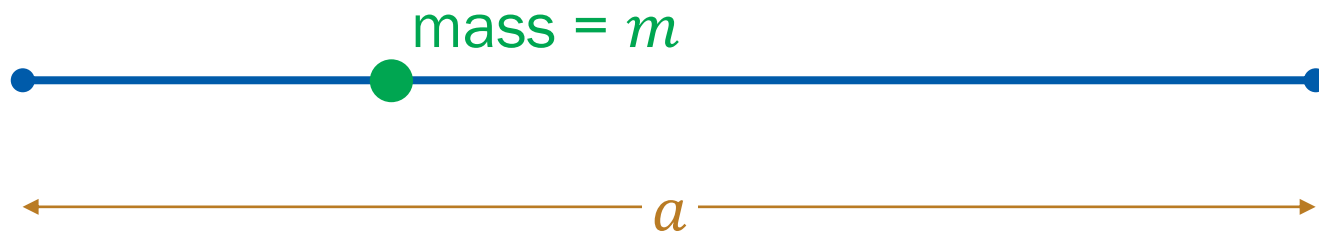
$$|T(t)|^2 = 1$$

$$|\Psi(x, y, z, t)|^2 = |\psi(x, y, z)|^2$$

Where Are We?



Particle in a 1D Box



$$V(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$

Schrödinger Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \hat{H} \Psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \mathcal{V}(x, t) \right] \Psi(x, t)$$

Since the potential is independent of time, we need to solve

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \mathcal{V}(x) \right] \psi(x) = E\psi(x), \quad -\infty < x < \infty$$

which is

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E\psi(x), \quad 0 \leq x \leq a$$

$$\psi(x) = 0, \quad \text{otherwise}$$

General Solution of the Equation

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\psi(x), \quad 0 \leq x \leq a$$

Solving this differential equation,

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

where $k = \sqrt{2mE}/\hbar$, and A and B are some constants.

Boundary Conditions

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

where $k = \sqrt{2mE}/\hbar$.

We know $\psi(0) = \psi(a) = 0$:

$$\psi(0) = B = 0$$

$$\psi(a) = A \sin(ka) = 0$$

For non-trivial solutions, $ka = n\pi$ (n : integer)

RS 3.1

M 3.5

System Energy

Since $k = \sqrt{2mE}/\hbar$ and $ka = n\pi$,

$$\frac{\sqrt{2mE}}{\hbar} = \frac{n\pi}{a}$$

$$E = \frac{1}{2m} \left(\frac{n\pi\hbar}{a} \right)^2 = \frac{h^2}{8ma^2} n^2$$

Wavefunction

$$\psi(x) = A \sin\left(\frac{n\pi x}{a}\right)$$

From the normalization condition,

$$\int_0^a |\psi(x)|^2 dx = |A|^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = |A|^2 \frac{a}{2} = 1$$
$$|A|^2 = 2/a$$

We “choose” $A = \sqrt{2/a}$, and

$$\psi(x) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right)$$

“Index” n

$$\psi_n(x) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right)$$

- $n = 0$: $\psi(x) = 0$ (trivial solution)
- $n < 0$: by letting $m = -n$,

$$\psi_n(x) = \sqrt{2/a} \sin\left(-\frac{m\pi x}{a}\right) = -\sqrt{2/a} \sin\left(\frac{m\pi x}{a}\right) = -\psi_m(x)$$

Hence, $|\psi_n(x)|^2 = |\psi_m(x)|^2$,

which means that n and $-n$ indicate the same state.

We only need to consider $n > 0$ (natural number)

Final Solution

- Index $n = 1, 2, 3, \dots$ (“quantum number”)
- Time-independent wavefunction

$$\psi_n(x) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right), \quad 0 \leq x \leq a$$

- Energy

$$E_n = \frac{h^2}{8ma^2} n^2$$

- Total wavefunction

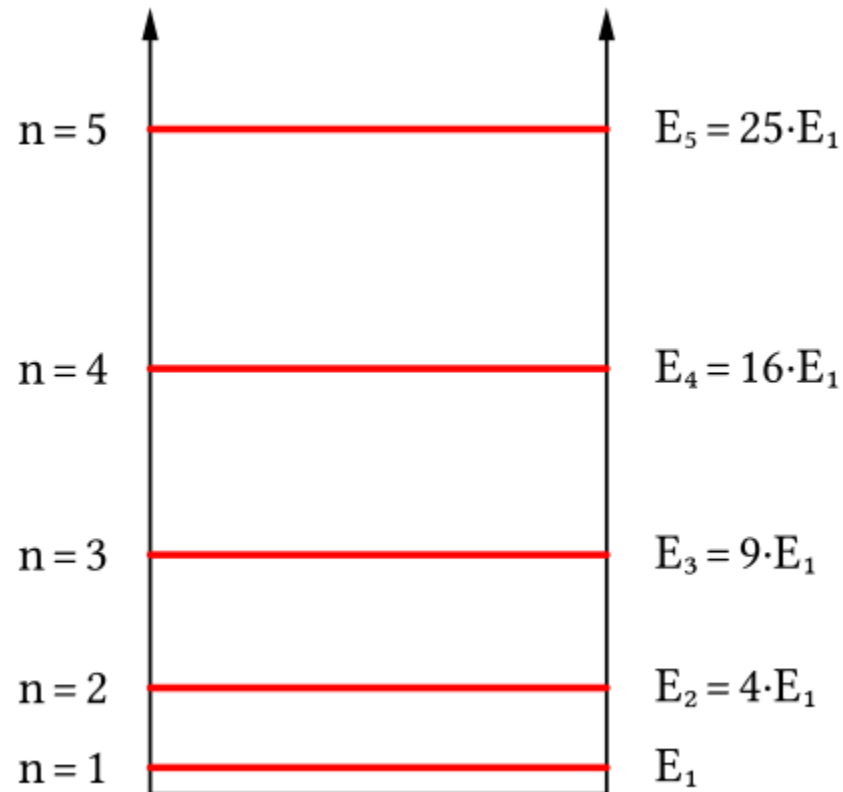
$$\Psi_n(x, t) = \psi_n(x)T_n(t) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}$$

RS 3.1

M 3.5

Energy Levels

$$E_n = \frac{h^2}{8ma^2} n^2$$



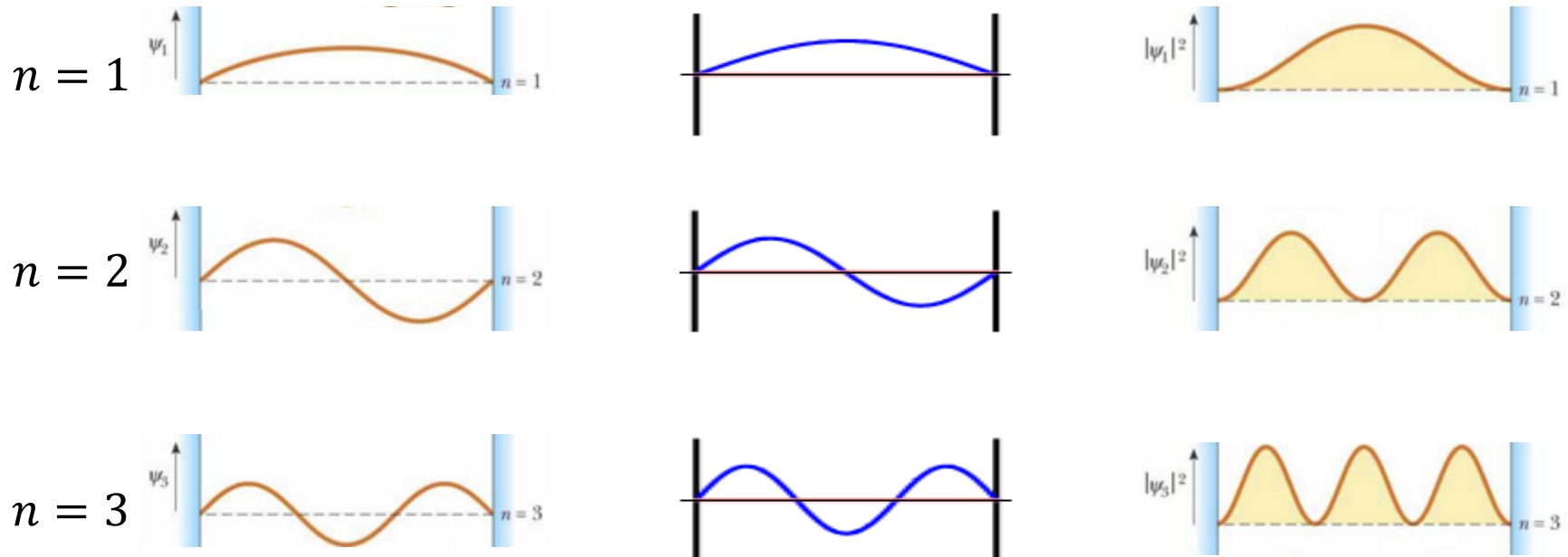
RS 3.1
M 3.5

Wavefunctions

$$\psi_n(x)$$

$$\Psi_n(x, t)$$

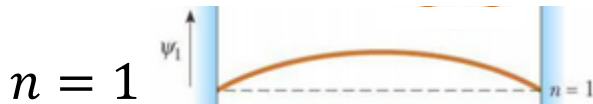
$$|\Psi_n(x, t)|^2$$



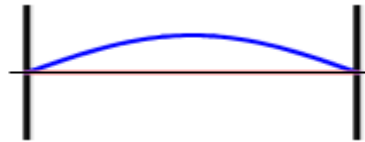
(Image: <https://web2.ph.utexas.edu/~coker2/index.files/ch35.htm>)

Example: $n = 1$ State

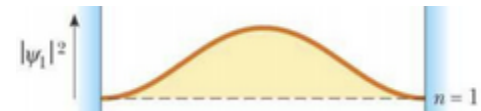
$$\psi_n(x)$$



$$\Psi_n(x, t)$$



$$|\Psi_n(x, t)|^2$$



If the system is at state $n = 1$, the wavefunction is

$$\psi_1(x) = \sqrt{2/a} \sin\left(\frac{\pi x}{a}\right)$$

Probability

The probability that the particle is found between $x = 0$ and $a/2$ is

$$\begin{aligned} P(0, a/2) &= \int_0^{a/2} |\Psi_1(x, t)|^2 dx = \int_0^{a/2} |\psi_1(x)|^2 dx \\ &= \int_0^{a/2} \left(\frac{2}{a}\right) \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{1}{2} \end{aligned}$$

Expectation Values

The expectation value of observable A is

$$\begin{aligned}\langle A \rangle &= \int_{-\infty}^{\infty} \psi_1^*(x) \hat{A} \psi_1(x) dx \\ &= \int_0^a \sqrt{2/a} \sin\left(\frac{\pi x}{a}\right) \hat{A} \left[\sqrt{2/a} \sin\left(\frac{\pi x}{a}\right) \right] dx\end{aligned}$$

Expectation Value of Position

$$\langle x \rangle = \frac{2}{a} \int_0^a \sin\left(\frac{\pi x}{a}\right) x \sin\left(\frac{\pi x}{a}\right) dx = \frac{2}{a} \int_0^a x \sin^2\left(\frac{\pi x}{a}\right) dx$$

$$\therefore \langle x \rangle = \frac{a}{2}$$



Expectation Value of Momentum

$$\begin{aligned}
 \langle p_x \rangle &= \frac{2}{a} \int_0^a \sin\left(\frac{\pi x}{a}\right) \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \sin\left(\frac{\pi x}{a}\right) dx \\
 &= \frac{2\hbar}{ia} \int_0^a \sin\left(\frac{\pi x}{a}\right) \left(\frac{\pi}{a} \right) \cos\left(\frac{\pi x}{a}\right) dx \\
 &= \frac{2\pi\hbar}{ia^2} \int_0^a \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{\pi x}{a}\right) dx \\
 &\quad \therefore \langle p_x \rangle = 0
 \end{aligned}$$

Variances

Similarly,

$$\langle x^2 \rangle = \frac{a^2}{3} - \frac{a^2}{2\pi^2}; \quad \langle p_x^2 \rangle = \frac{h^2}{4a^2}$$

which leads to

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{a^2}{3} - \frac{a^2}{2\pi^2} - \left(\frac{a}{2}\right)^2 = \frac{a^2}{12} - \frac{a^2}{2\pi^2}$$

$$\sigma_{p_x}^2 = \langle p_x^2 \rangle - \langle p_x \rangle^2 = \frac{h^2}{4a^2}$$

Uncertainty Principle

Now,

$$\sigma_x^2 \sigma_{p_x}^2 = \left(\frac{a^2}{12} - \frac{a^2}{2\pi^2} \right) \frac{h^2}{4a^2} = h^2 \left(\frac{1}{48} - \frac{1}{8\pi^2} \right) = 0.00817h^2$$

$$\sigma_x \sigma_{p_x} = 0.090h$$

The uncertainty principle states that

$$\sigma_x \sigma_{p_x} \geq h/4\pi = 0.080h$$